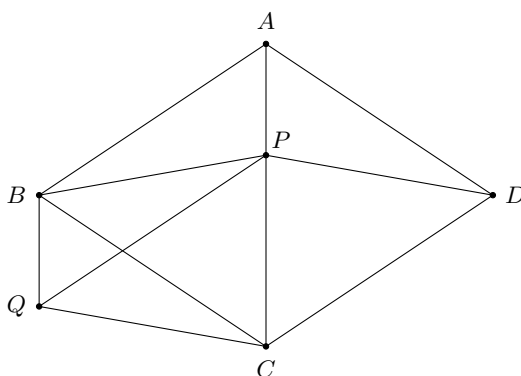


**Problem 29.** Given a rhombus  $ABCD$ , find the locus of the points  $P$  lying inside the rhombus and satisfying  $\angle APD + \angle BPC = 180^\circ$ .

**Solution.** Let  $Q$  be a point such that  $PQCD$  is a parallelogram. Then  $ABQP$  is also a parallelogram. From the equality  $\angle APD + \angle BPC = 180^\circ$  it follows that  $\angle BQC + \angle BPC = 180^\circ$ , so the points  $B, Q, C, P$  lie on a common circle. Therefore,  $\angle PBC = \angle PQC = \angle PDC$ , and since  $BC = CD$ , we obtain that  $\angle CPB = \angle CPD$  or  $\angle CPB + \angle CPD = 180^\circ$ . Hence, the point  $P$  lies on the segment  $AC$  or on the segment  $BD$ .

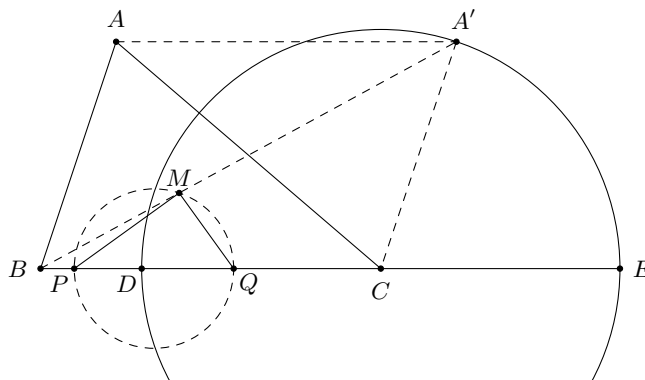
Conversely, any point  $P$  lying on the diagonal  $AC$  satisfies the equation  $\angle BPC = \angle DPC$ . Therefore,  $\angle APD + \angle BPC = 180^\circ$ . Analogously, we show that the last equation holds if the point  $P$  lies on the diagonal  $BD$ .



□

**Problem 30.** Let the medians of  $\triangle ABC$  meet at  $M$ . Let  $D$  and  $E$  be distinct points on the line  $BC$  such that  $DC = CE = AB$ , and let  $P$  and  $Q$  be points on the segments  $BD$  and  $BE$ , respectively, such that  $2BP = PD$  and  $2BQ = QE$ . Determine, with reasons,  $\angle PMQ$ .

**Solution.** Draw the parallelogram  $ABCA'$ . Then  $M$  lies on  $BA'$  since the diagonals of a parallelogram bisect each other. Also, we have  $BM = \frac{1}{3}BA'$  by similar triangles since the centroid  $M$  cuts a median in the ratio  $2 : 1$ . So  $M$  is on the homothetic image (centre  $B$ , dilation  $1/3$ ) of the circle with centre  $C$  and radius  $AB$ . Since this circle meets  $BC$  at  $D$  and  $E$ , its image meets  $BC$  at  $P$  and  $Q$ . So  $\angle PMQ = \angle DA'E = 90^\circ$ . The last equality holds since  $CD = CE = AB = CA'$ .



□

**Problem 31.** Show that  $n^n + (n+1)^{n+1}$  is composite for infinitely many positive integers  $n$ .

**Solution.** We will show that for any positive integer  $n \equiv 4 \pmod{6}$  the number  $n^n + (n+1)^{n+1}$  is divisible by 3 and hence composite since it is larger than 3. Indeed, for any such  $n$  we have  $n \equiv 1 \pmod{3}$  and hence  $n^n + (n+1)^{n+1} \equiv 1^n + 2^{n+1} = 1 + 2^{n+1} \pmod{3}$ . Moreover, the exponent  $n+1$  is odd, which implies that  $2^{n+1} \equiv 2 \pmod{3}$ . It follows that  $n^n + (n+1)^{n+1} \equiv 0 \pmod{3}$ , as claimed. □

**Problem 32.** Two persons play the following game with integers. The initial number is  $2019^{2019}$ . The players move in turns. Each move consists of subtraction of an integer between 1 and 2018 inclusive, or division by 2019, rounding down to the closest integer when necessary. The player who first obtains a non-positive integer wins. Which player has a winning strategy?

**Solution.** Let  $n = 2019^a m$  be the initial number where  $a \geq 0$  and  $m$  is not divisible by 2019. We claim that the second player wins if and only if  $a$  is odd.

The base case  $n = 1$  is trivial. Assume all smaller cases are done. Consider the case  $n \geq 2$ . If  $a = 0$ , let  $r$  be the remainder obtained when  $n$  is divided by 2019. Then one of the numbers  $n - r$  and  $\frac{n - r}{2019}$  is of the form  $2019^b k$  where  $b$  is odd and  $k$  is not divisible by 2019. So the first player  $X$  can either subtract  $r$  or divide by 2019 and round down to get such a number. Then  $X$  has a winning strategy from the inductive hypothesis (note that the roles of the players are changed). Similarly, if  $a \geq 2$  is even, then  $X$  can divide the number by 2019 to get a number for which  $X$  has a winning strategy.

If  $a$  is odd,  $X$  cannot win in any case. Indeed, if  $X$  subtract a number, then the remaining number is not divisible by 2019 and so the second player can win. If  $X$  divides the number by 2019, then the exponent of 2019 becomes even, which means the second player can win. So the claim holds by induction.

When the initial  $n$  is  $2019^{2019}$ , the second player will win.

□